

THIERRY RUGIRA

Data & AI Engineering | Systems & Operational Automation

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PROFILE

Data and AI engineering practitioner building reliable data systems and operational automations in Rwanda. Work spans healthcare utilization, disease-surveillance monitoring, environmental analysis and support workflows. Focused on data quality, maintainability and practical AI within real resource constraints.

EXPERIENCE

SAND Technologies | Kigali, Rwanda

Employee | December 2024 - Present | Work focus: data & AI engineering

- Build Python / Prefect data workflows and PostgreSQL-backed systems, with scheduled ingestion, data-quality checks, backfills and missing-detail recovery tools.
- Develop operational automation, system integrations, FastAPI interfaces and practical AI solutions; work across scheduled jobs, webhook handoffs and data validation.
- Contribute to eBuzima healthcare utilization pipelines in a data context of approximately 10,500 rows per day across roughly 450 facilities.
- Automate REMA / PurpleAir analysis: combine sensor data, assess completeness, verify averages and generate reporting tables and visualizations.

SELECTED SYSTEMS & PRODUCT WORK

Impuruza | Disease-surveillance monitoring

Contribute to an implemented DHIS2-to-Zammad workflow: scheduled Python monitoring, metadata enrichment and signal-age checks, with webhook delivery through n8n to create follow-up tickets.

Issue-tracking automation | Airtable to NocoDB

Built migration tooling with field and date mapping, schema checks, duplicate detection, dry-run validation and batched API writes. Complements broader scheduled processing and data-recovery work.

Waka gym platform | Prototype in development

Systems and product prototyping for gym operations, connecting operational workflows with backend structure and user experience.

TECHNICAL SKILLS

Data: Python, SQL, PostgreSQL, Prefect, dbt, data quality, recovery workflows

Automation & integration: Scheduled jobs, webhooks, FastAPI, DHIS2, Airtable / NocoDB

Systems: AI solutions, product prototyping, n8n / Zammad ticket workflows

EDUCATION

African Leadership University

Bachelor of Software Engineering (BSE), in progress

Current studies: Year 2 | Expected completion: January 2028